A Systematic Review of Machine Learning and Deep Learning Approaches for Credit Card Fraud Detection

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Abstract*: The increasing sophistication of payment card fraud needs enhanced artificial intelligence (AI) treatments. This systematic review critically assesses the landscape of Machine Learning (ML) and Deep Learning (DL) applications in credit card fraud detection from 2021 to 2024 (77 studies), focusing on unsolved AI obstacles and potential for advancement. While popular algorithms such as Random Forest and Logistic Regression continue to thrive, and deep learning methods such as CNNs and LSTMs gain traction, our analysis reveals critical gaps in addressing fundamental AI problems such as robustness against adversarial attacks, effective concept drift management, comprehensive missing data imputation strategies, and the untapped potential of emerging AI paradigms such as Large Language Models (LLMs) and advanced optimization techniques. This paper is more than just a method review; it lays out a strategic AI research roadmap for future advances in fraud detection, highlighting the importance of explainable, adaptive, and scalable AI solutions to manage shifting risks and practical implementation challenges.

Keywords: Credit Card Fraud Detection, Machine Learning, Deep Learning, AI-based Optimization, Systematic Review.

I. INTRODUCTION

The rapid increase in digital financial activities, accelerated by the recent global health crisis, has led to a substantial increase in the volume of electronic transactions. A recent study stated that in the second quarter of 2021, Mastercard and Visa issued 1,131 million and 1,156 million cards, respectively [1]. However, this surge has introduced significant challenges, particularly concerning illicit activities involving payment cards. Such fraudulent practices have become a major concern for financial organizations worldwide. Global losses attributed to payment card fraud have increased significantly over the past decade. It is anticipated that the problem will become more severe in the future. Projections indicate that worldwide losses from payment card fraud could reach 49.32 billion USD by 2030 [2]. This consistent upward trend highlights the increasing complexity of fraudulent activities and the difficulties in effectively combating them.

A report indicated that financial losses due to credit and debit cards are among the most significant sources of losses in the financial sector. Therefore, developing efficient credit card fraud-detection systems is necessary to mitigate these losses. This sharp increase is mainly driven by volume of card transactions and the expansion of digital payment platforms, highlighting the urgent need for more advanced fraud prevention techniques [1]. Traditional methods for detecting payment card fraud typically rely on rule-based systems and statistical analysis. These systems apply predefined criteria, such as flagging transactions that exceed a specific value or originate from unusual locations. While these methods are straightforward and easy to implement, they have notable limitations. Their rigidity prevents them from adapting to evolving fraud tactics, often resulting in high false-positive rates that inconvenience legitimate customers and increase the need for manual reviews. Furthermore, these systems are often reactive, identifying fraud only after it has occurred rather than proactively preventing it, which can lead to financial losses and reputational damage [3]. The complex nature of modern fraud requires more sophisticated solutions, such as machine learning (ML). Machine learning algorithms have revolutionized the field of fraud detection by offering automated data analysis and advanced pattern recognition capabilities [4]. These systems can process vast quantities of data and identify subtle patterns that traditional methods might overlook. Additionally, ML models continuously learn and improve as new data becomes available, enabling more accurate and proactive fraud detection. By leveraging ML, financial institutions can significantly improve fraud detection and prevention, ultimately improving security and customer satisfaction [3, 5].

Deep Learning (DL) represents a significant advancement in credit card fraud detection, offering superior pattern recognition capabilities compared to traditional rule-based and machine learning models. DL models, such as recurrent neural networks (RNNs) and autoencoders, can automatically learn complex transaction patterns, enhancing real-time anomaly detection. Moreover, DL models can scale effectively with increasing transaction volumes, making them particularly suitable for large financial institutions and payment processors that handle millions of transactions daily [6]. The digital transformation of finance has inadvertently amplified the threat of credit card fraud, evolving into a sophisticated, dynamic, and economically damaging challenge. Detecting such fraud is not merely a classification problem but a complex Artificial Intelligence (AI) challenge characterized by extreme data imbalance, continuous concept drift due to evolving fraudster tactics, the necessity for real time processing, and the critical demand for model interpretability. While existing literature offers reviews on ML/DL applications in this domain, a comprehensive systematic analysis that critically evaluates the AI methodologies, identifies specific AI limitations, and provides a forward-looking AI research roadmap remains largely unexplored. This paper addresses this gap by conducting a rigorous systematic review of 77 studies published between 2021 and 2024. Our objective is not just to summarize prevalent ML/DL techniques, but to analyze their inherent AI strengths and weaknesses, uncover significant omissions such as the underutilization of advanced optimization and emerging AI paradigms like Large Language Models (LLMs), and ultimately, to chart a clear course for future AI research to develop more robust,

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adaptive, and explainable fraud detection systems. The subsequent sections outline the methodology, present key findings, and discuss their implications for future research directions and practical applications.

II. RESEARCH QUESTIONS

The study utilized the PICO framework, as shown in Table 1: Population, Intervention, Comparison, Outcomes along with Language and Timing criteria, to define relevant keywords and construct precise search strings for addressing research questions.

Table 1: PICO Criteria

Population	Financial institutions specifically banks
Intervention	Using artificial intelligent technologies in credit card fraud detection
Comparison	In this investigation, no comparative intervention is anticipated.
Outcomes	Enhance fraud detection systems in banks using machine and deep learning techniques
Timing	4 years period
Language	Articles were reported in English

The review comprises two main parts: a systematic literature review and a bibliometric analysis, both aimed at achieving the study's objectives. Bibliometric analysis serves as the initial phase of the review and addresses the following questions:

1) What is the distribution of studies per year?

2) Which countries have contributed the most to this topic?

3) What are the leading journals publishing studies on machine learning and deep learning approaches for credit card fraud detection?

The second phase, the systematic review, focuses on the following research questions:

1) How many credit card fraud detection articles utilize machine learning, deep learning, or both?

2) What fraud detection algorithms are commonly used in this area?

3) How many studies discuss supervised, unsupervised, or semi-supervised techniques?

4) How frequently are class imbalance issues addressed?

5) How frequently is the missing data problem addressed? 6) How many articles incorporate optimization techniques?

7) What are the top 10 AI-based optimization techniques used in credit card fraud detection, and how do they compare in terms of implementation frequency? 8) What gaps remain in this area?

III. RESEARCH STRUCTURE

The review is structured to provide a comprehensive analysis of AI techniques in credit card fraud detection. It begins with an **Introduction**, which establishes the significance of the topic, outlines the objectives, and situates the research within the broader financial and technological landscape. The research problem is clearly defined, highlighting its importance and linking it to existing literature. This is followed by Re The review is structured to provide a comprehensive analysis of AI techniques in credit card fraud detection. It begins with an **Introduction**, which establishes the significance of the topic, outlines the objectives, and situates the research within the broader financial and technological landscape. The research problem is clearly defined, highlighting its importance and linking it to existing literature. This is followed by **Research Questions**, which frame the investigation and ensure a focused exploration of the topic. The **Theoretical Background** offers an overview of relevant theories and concepts, sitting the study within its academic context. Additionally, the **Meta-Survey** summarizes trends in AI-based fraud detection and identifies commonly used methods and techniques. The **Review Methodology** details the search strategy, inclusion and exclusion criteria, and data extraction and synthesis methods, ensuring quality and rigor in the analysis. The **Results** section synthesizes key findings, highlighting emerging trends and research gaps. Finally, the **Conclusion** consolidates the main findings, discusses their implications, acknowledges limitations, and suggests future research directions. This structured approach ensures a coherent and detailed examination of the topic.

IV. THEORETICAL BACKGROUND

A. Financial fraud overview

Financial fraud refers to any deliberate act of deception or manipulation intended to gain an unauthorized financial advantage, often at the expense of individuals, businesses, or institutions. Financial fraud encompasses a wide range of activities, including identity theft, credit card fraud, insurance fraud, money laundering, embezzlement, and phishing scams. Perpetrators exploit vulnerabilities in financial systems, digital platforms, and human behavior to carry out fraudulent activities. Financial fraud poses significant economic, legal, and reputational risks, highlighting the need for robust security measures, regulatory compliance, and public awareness to effectively detect and prevent such crimes.

1) Insurance fraud

Insurance fraud involves the deliberate deception or misrepresentation by individuals or organizations to obtain financial benefits or advantages to which they are not entitled. It can occur at various stages of the insurance process, including application, claim submission, or policy renewal. Common types of insurance fraud include falsifying claims, staging accidents, inflating damages, and providing false information to secure lower premiums. Perpetrators may range from policyholders and third-party claimants to professionals such as doctors or contractors who collaborate to defraud insurers. Insurance fraud not only causes financial losses for insurance companies but also leads to higher premiums for honest policyholders, underscoring the need for effective fraud detection systems and strict regulatory oversight [7].

2) Bank fraud

Bank fraud is a broad category of financial crime involving illegal activities designed to deceive banks or financial institutions to gain unauthorized access to funds, assets, or services. This includes actions such as falsifying documents, forging signatures, manipulating account balances, or transferring funds without the account holder's knowledge or consent. It can also involve complex schemes like identity theft or wire fraud, where criminals use fake identities or hijack legitimate accounts to steal money. One of the most common forms of bank fraud is

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credit card fraud, which occurs when individuals gain unauthorized access to someone's credit card information and use it for personal gain [8].

3) Credit card fraud

Credit card fraud can take several forms. Card-not-present fraud occurs when thieves use stolen credit card details to make online or phone transactions without physically possessing the card. This type of fraud is often facilitated by data breaches, phishing attacks, or the sale of stolen credit card information on the dark web. In contrast, cardpresent fraud involves the physical theft or cloning of a credit card, enabling criminals to conduct fraudulent transactions by mimicking the legitimate cardholder. Fraudsters may also use skimming, a technique where a small device copies information from a card's magnetic stripe at ATMs or point-of-sale terminals. Additionally, they may exploit fraudulent applications to open new accounts or obtain loans under false pretenses [9].

4) Taxation fraud

Tax fraud involves illegal manipulation of a business's structure or financial records to reduce tax liability. Unlike tax evasion, which relies on legal methods to minimize taxes, tax fraud entails deliberate actions to mislead or deceive tax authorities. This includes falsifying income, inflating expenses, or concealing assets to evade tax payments. Tax fraud is driven by the intent to violate the law and often exploits weaknesses in the tax system, inadequate enforcement, or legislative loopholes. Its consequences are severe, leading to significant public revenue losses and straining government resources. Despite ongoing efforts to combat it, tax fraud remains a persistent issue, particularly during economic crises when its impact on public finances becomes more pronounced [10].

5) Corporate fraud

Corporate fraud refers to deceptive practices carried out by individuals or groups within a company to secure an unfair financial gain, often at the expense of investors, employees, customers, or stakeholders. It can take various forms, including financial statement manipulation, embezzlement, insider trading, bribery, accounting fraud, and misappropriation of company assets. Corporate fraud often involves senior executives or employees exploiting their positions of trust to conceal financial discrepancies, inflating profits, or cover operational losses. Such fraudulent activities can severely damage a company's reputation, lead to significant financial losses, and result in legal penalties, underscoring the need for strong corporate governance, transparency, and regulatory compliance to prevent and detect fraud. *6) Mortgage fraud*

Mortgage fraud involves deceptive practices in real estate transactions to unlawfully obtain financial gain. It falls into two main categories: fraud-for-property and fraud-forprofit. Fraud-for-property occurs when borrowers, often aided by real estate professionals, falsify income or assets to secure loans they cannot afford. In contrast, fraud-forprofit schemes, driven by greed, typically involve industry insiders inflating property values to extract fraudulent funds. These schemes often rely on identity theft, forged documents, and misrepresentations during the mortgage lending process [11].

B. Artificial Intelligence technologies overview

Artificial intelligence (AI) is a broad field of computer science focused on developing systems capable of performing tasks that typically require human intelligence. These tasks include learning, problem-solving, decision-making, and understanding language and perception. AI technologies encompass machine learning, deep learning, natural language processing, computer vision, robotics, and expert systems, each enabling machines to analyze data, recognize patterns, make decisions, and automate tasks across various fields.

1) Machine Learning (ML)

Machine learning (ML) is a subfield of AI that enables systems to learn from data without explicit programming. ML algorithms build a model based on sample data, known as training data, to make predictions or decisions. This learning process allows systems to enhance their performance over time as they are exposed to more data [12].

2) Deep Learning (DL)

Deep learning (DL) is a specialized subfield of machine learning that utilizes artificial neural networks with multiple layers (deep neural networks) to analyze data. DL models automatically extract complex features from raw data, making them particularly effective for tasks involving unstructured data such as images, text, and speech [13].

3) Data Mining

Data mining is the process of discovering patterns, anomalies, and valuable insights from large datasets. It leverages techniques from statistics, machine learning, and database systems to extract knowledge and transform raw data into actionable information for decision-making and analysis.

4) Expert Systems

Expert systems are AI-driven applications designed to replicate human expertise in specific areas. They utilize a knowledge base and a set of rules to analyze problems and provide solutions or recommendations. Expert systems are widely applied in fields such as diagnosing diseases in healthcare, detecting fraud in finance, and troubleshooting technical issues in engineering [14].

5) Computer Vision

Computer vision is a branch of artificial intelligence focused on enabling machines to process and interpret visual information from the world. It encompasses tasks such as image recognition, object detection, and video analysis. Computer vision is widely applied in self-driving cars for object detection, medical imaging for diagnosing conditions, and security systems for facial recognition [15].

6) Chatbots

Chatbots are conversational AI systems designed to interact with users in a natural, human-like manner. They can process text or voice inputs and generate responses based on programmed scripts or AI-driven algorithms. They are commonly used in customer service, virtual assistants, and e-commerce platforms to answer inquiries, provide support, and assist with tasks such as booking appointments or processing orders [16].

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V. META-SURVEY

This section provides an overview of prior reviews on credit card fraud detection using AI techniques as illustrated in Table 2. O. A. Bello explores the application of deep learning, particularly neural networks and anomaly detection, in cyber financial fraud detection. The study examines the use of CNNs and RNNs for detecting fraud patterns and employs autoencoders and GANs for anomaly detection in the absence of labeled data. It highlights key challenges, including data quality and computational resources constraints, while emphasizing the necessity of continuous model training. The objective is to demonstrate how deep learning can enhance fraud detection accuracy and efficiency in the financial sector [3]. The systematic review by Suraya Nurain Kalid explores credit card fraud and payment default challenges, focusing on **imbalanced and overlapping classes**. It reviews **machine learning solutions**, recommending **deep learning, ensemble methods, and sampling techniques** to improve detection. The study also stresses the use of **evaluation metrics** that assess performance separately for fraud, defaults, and normal transactions. [5]. The systematic review by Rui Miguel Dantas explores the growing issue of credit card fraud, particularly in the context of cybercrimes. It examines a range of articles from leading databases to identify key gaps and research opportunities in the detection of fraud using machine learning. The review highlights the effectiveness of both supervised and unsupervised machine learning techniques in combating cybercrime within the credit card industry. While the field has gained prominence over the past decade, the study emphasizes the need for further advancements in both quantitative and qualitative research to enhance fraud detection methods [20].

The objective of our study is to conduct a comprehensive bibliometric analysis of previous research on credit card fraud detection, identifying key trends, prominent research areas, and the evolution of the field. We focus on the most used machine learning algorithms and deep learning techniques for fraud detection, evaluating their effectiveness and limitations. Additionally, we examine studies that address two major data challenges missing data and class imbalance highlighting the strategies employed to mitigate these challenges. Finally, this study aims to uncover research gaps, particularly in the areas of handling missing data, addressing class imbalance, and improving the overall effectiveness of machine learning methods in credit card fraud detection.

Table 2: Comparison of other related studies in the same domain: (✓: YES, ✗: NO)

Authors	Year	Objective of study	Methodology				AI Technology		
			Bibliometric	Systematic Review	Meta Analysis	Content Analysis	Machine Learning	Deep Learning	Data Mining
Matin et al. **[17]**	2021	Review literature on financial statement fraud detection using ML/DM techniques, datasets, and research gaps.	✗	✓	✓	✓	✓	✗	✓
Marco Sánchez et al **[18]**	2021	Analyze literature on fraud detection using ML and fraud triangle/diamond theories to identify gaps and future directions.	✓	✓	✗	✗	✓	✓	✓
Kalid et al. **[5]**	2024	Conduct a review to compare credit card datasets, machine learning techniques, and evaluation metrics.	✗	✓	✓	✓	✓	✓	✓
Eyad et al. **[19]**	2023	This study explores machine learning and deep learning techniques for credit card fraud detection, highlighting their advantages over traditional methods.	✗	✓	✗	✓	✓	✓	✗
Rui, et al **[20]**	2022	The paper reviews machine learning in financial cybercrime detection, focusing on credit card fraud, performance metrics, datasets, and research gaps.	✓	✗	✗	✓	✓	✗	✗
O. A. Bello, et al **[3]**	2022	The paper aims to analyze techniques used for credit card fraud detection, exploring their effectiveness, limitations, and possible improvements	✗	✓	✗	✗	✓	✗	✓
Our review	2025	This study conducts a bibliometric analysis, identifying key trends, methods, and challenges	✓	✓	✗	✓	✓	✓	✗

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VI. Review Methodology

This section outlines the approach used to conduct the review. A systematic literature review methodology is employed to investigate credit card fraud detection techniques. As illustrated in Fig. 1, the process begins with a **Search Strategy**, which defines the targeted keywords to refine the search context. The search is conducted within the **Scopus database**. Next, the **Studies Selection** phase establishes criteria for identifying, screening, and selecting relevant studies. Following this, the **Data Analysis** phase applies bibliometric and content analysis to systematically review and evaluate the identified literature. Finally, the **Results** synthesize insights from the bibliometric analysis and present the conclusions of the systematic review. This structured approach ensures a comprehensive evaluation of the field.

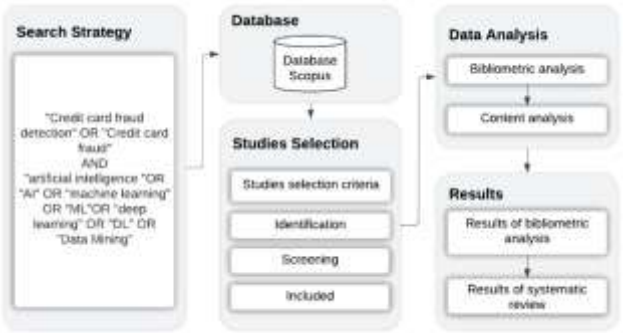


Fig. 1. Systematic review methodology

A. Search Strategy

A comprehensive search for publications on credit card fraud detection using AI techniques was conducted on December 20, 2024, utilizing the Scopus database. The advanced search functionality in Scopus enabled the use of Boolean operators such as "OR" and "AND" to refine the results.

The search parameters were defined as follows:

1) journal articles and conference papers, 2) published between 2021 and 2024, 3) written in English, 4) finalized and at the publication stage, and 5) within the field of computer science.

Detailed descriptions of the search parameters are provided below.

TITLE-ABS-KEY("credit card fraud detection" OR "Credit card fraud" AND ("artificial intelligence "OR "AI" OR "machine learning" OR "ML"OR "deep learning" OR "DL" OR "Data Mining")) AND PUBYEAR > 2020 AND PUBYEAR < 2025 AND (EXACTKEYWORD("Fraud Detection") OR EXACTKEYWORD("Credit Card Fraud")) AND SUBJAREA("COMP") AND PUBYEAR > 2020 AND PUBYEAR < 2025 AND (LIMIT-TO (PUBSTAGE,"final") OR LIMIT-TO (PUBSTAGE,"English")) AND (LIMITTO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE,"cp"))

B. Database Sources

Scopus was selected as the primary database for this systematic review because it offers broad coverage of scientific publications, including conference proceedings and journal articles. It is recognized as a reliable platform for accessing high-impact research across various academic disciplines. Its advanced search features, such as the use of logical operators, align with the requirements of the systematic review methodology. Additionally, Scopus provides robust filtering and bibliometric analysis capabilities, ensuring a comprehensive and structured approach to the literature review.

C. Study Selection Criteria

Study selection criteria define the specific guidelines used to determine which studies are included or excluded in a research review, ensuring relevance to the research objective. Each paper was evaluated based on the inclusion and exclusion criteria outlined below:

1) Eligibility Criteria

A paper was included if its title contained “credit card fraud detection" OR "Credit card fraud" AND "artificial intelligence "OR "AI" OR "machine learning" OR "ML"OR "deep learning" OR "DL" OR "Data Mining", The same criteria applied to the abstract. If the title and abstract were insufficient for determining relevance, the full text of the article was reviewed to assess whether it met these requirements.

2) Inclusion/Exclusion Criteria

In addition to the above, the articles’ relevancy was filtered using the following criteria set:

- **Topic Relevance**: Papers must address credit card fraud detection using techniques such as AI, ML, DL, or data mining.
- **Keywords**: The research must include specific keywords such as "Fraud Detection" or "Credit Card Fraud."
- **Publication Year**: Papers published between 2021 and 2024 are considered.
- **Language**: Only articles written in English are included.
- **Subject Area**: The study must fall under the field of computer science.
- **Document Type:** The analysis includes journal articles ("ar") and conference papers ("cp").
- **Publication Stage:** Papers must be at the final publication stage.
- **Availability**: Publications must be accessible via the Scopus database and available for full-text download.

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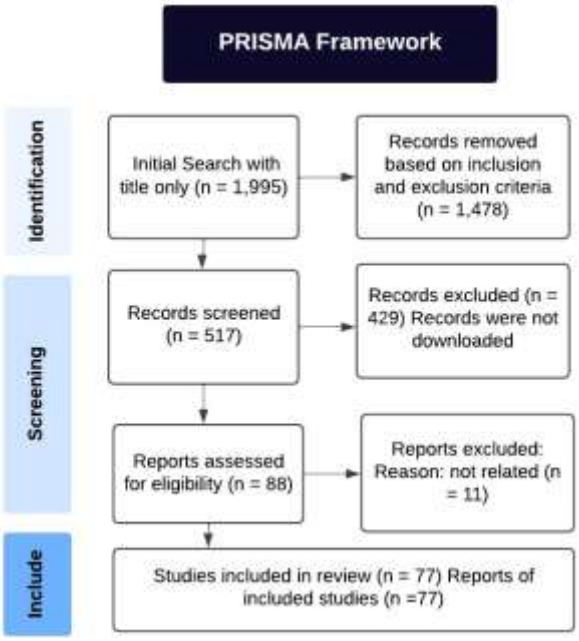


Fig. 2. Flowchart detailing the data collection, assessment, and inclusion processes

a) Identification

The research process began with an extensive search of the Scopus database, yielding an initial pool of 1,995 records. This broad search aimed to capture all potentially relevant studies within the scope of the research scope. The next step involved applying inclusion and exclusion criteria to ensure relevance, leading to the removal of 1,478 records that did not meet the predefined conditions. As a result, the dataset was narrowed to 517 records for further analysis.

b) Screening

After the initial identification phase, as shown in Fig. 2, the remaining 517 records underwent a detailed screening process to assess their relevance and suitability for inclusion. At this stage, 429 records were excluded, primarily due to unavailability for download, preventing further analysis. The 88 reports that passed the screening were then rigorously evaluated for alignment with the study's objectives, leading to the exclusion of 11 reports that were deemed unrelated to the research focus.

c) Included

Ultimately, 77 studies were deemed relevant and of sufficient quality for inclusion in the final review. These studies form the foundation of research analysis, offering valuable insights and data to address the research questions

d) Data synthesis and analysis

To gain a comprehensive understanding of our subjects, we integrated bibliometric and content analysis. This dual approach leveraged the strengths of both methods, allowing us to benefit from the broad insights of bibliometric analysis while delving into the in-depth knowledge provided by content analysis. Bibliometric analysis systematically assessed the research landscape. By examining publication and citation data, identifying key studies, trends, and patterns in credit card fraud detection research. It helped map research areas, uncover knowledge gaps, assess research impact, and guide future research efforts. Meanwhile, content analysis systematically examined qualitative data from selected studies, offering an in-depth understanding of research findings, identifying key themes and concepts, and uncovering underlying meanings. This approach provided rich qualitative data to complement the insights gained from bibliometric analysis.

VII. Results

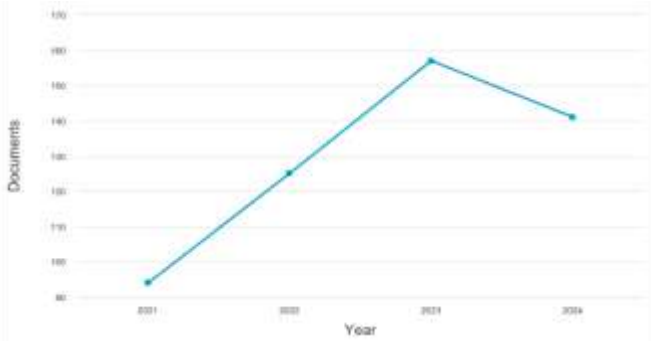
A. Bibliometric Analysis

Bibliometric analysis provides a structured approach to mapping the intellectual landscape of AI techniques in credit card fraud detection by analyzing publications and citations.

RQ1: What is the distribution of studies per year?

Through the analysis of 517 included papers, Fig. 3 illustrates the annual distribution of studies from 2021 to 2024 reveals a clear upward trend, reflecting a significant increase in research activity on AI-driven fraud detection. The number of studies rose from 94 in 2021 to 125 in 2022, followed by a further increase to 157 in 2023. Although there was a slight decline to 141 studies in 2024, the overall volume remains higher than in previous years. This trend highlights the growing interest in AI-driven solutions for fraud detection, likely driven by the escalating sophistication of fraudulent activities and the need for more advanced detection mechanisms. The consistent growth in research output underscores the dynamic nature of this field and the ongoing need for innovation to address emerging challenges in credit card fraud.

Fig. 3. Distribution of studies per year



RQ2: Which countries have contributed the most to this topic?

Fig. 4 highlights the geographic distribution of research contributions, with India leading as the leading contributor to AI-driven credit card fraud detection research. This dominance can be attributed to India's rapid digital transformation, a vast technology workforce, and the increasing adoption of digital payment systems, necessitating advanced fraud detection mechanisms. China and the United States follow closely, reflecting their strong technological infrastructure, substantial investment in AI research, and the high volume of digital transactions in these regions.

Saudi Arabia and Morocco also demonstrate notable contributions, reflecting a growing interest in leveraging AI to combat fraud in emerging digital economies. The inclusion of countries like Bangladesh, Iraq, and Egypt in the top ten further underscores that credit card fraud detection is a global concern, with research activity extending to regions undergoing rapid digitalization. However, the relatively lower contributions from the United Kingdom and Australia may indicate differing research priorities or a more mature fraud detection ecosystem in these regions. This geographic

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diversity underscores the worldwide significance of AI in fraud prevention and highlights the need for collaborative, cross-border research efforts to effectively address this universal challenge.

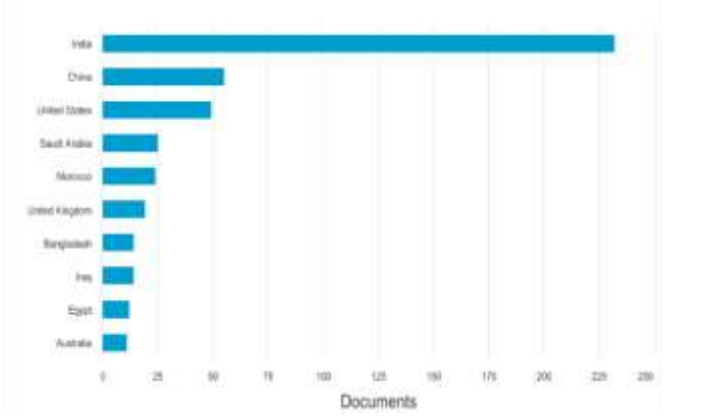


Fig. 4. Countries contribution in this topic

RQ3: What are the leading journals publishing studies on machine learning and deep learning approaches for credit card fraud detection?

The analysis of publication trends in machine learning and deep learning for credit card fraud detection reveals a dynamic interplay between research volume and academic influence. Fig. 5, which highlights the top five journals by document contribution, shows a notable upward trend for IEEE Access, peaking in 2024, indicating a considerable increase in research output in this domain. A similar trend is observed with Lecture Notes in Networks and Systems, which experienced substantial growth, culminating in 2023. Conversely, Advances in Intelligent Systems and Computing maintained a stable publication rate, reflecting a consistent and steady contribution to the field. This disparity in publication volume underscores the evolving nature of research activity within this specialized domain, demonstrating shifting academic priorities and emerging research directions.

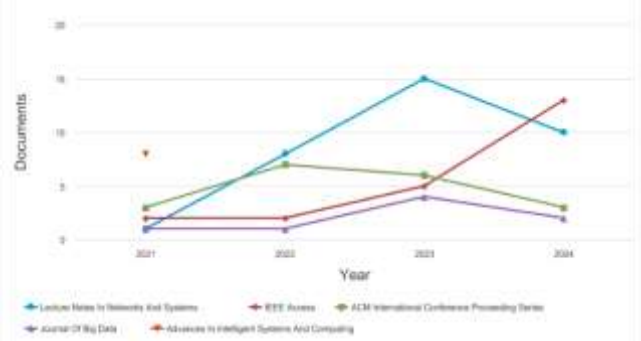


Fig. 5. Distribution of studies per year

Fig. 6 and the accompanying analysis of CiteScore trends provide a complementary perspective on the quality and impact of research in AI-driven credit card fraud detection. IEEE Access, despite its high publications volume, saw a remarkable rise in CiteScore from 2017 onwards, peaking in 2022. This sharp increase indicates that the journal has been attracting high-quality research in recent years, reinforcing its growing influence. Meanwhile, Lecture Notes in Networks and Systems and Advances in Intelligent Systems and Computing maintained relatively stable CiteScores, reflecting a robust reputation and stable academic standing within the field. Additionally, the upward CiteScore trends in ACM International Conference Proceedings Series and Journal of Big Data highlight a growing emphasis on impactful publications within these journals, further solidifying their role in advancing research in this area. These insights collectively provide a comprehensive view of the research landscape, underscoring the significant contributions of these journals to the advancement of AI-based credit card fraud detection methods.

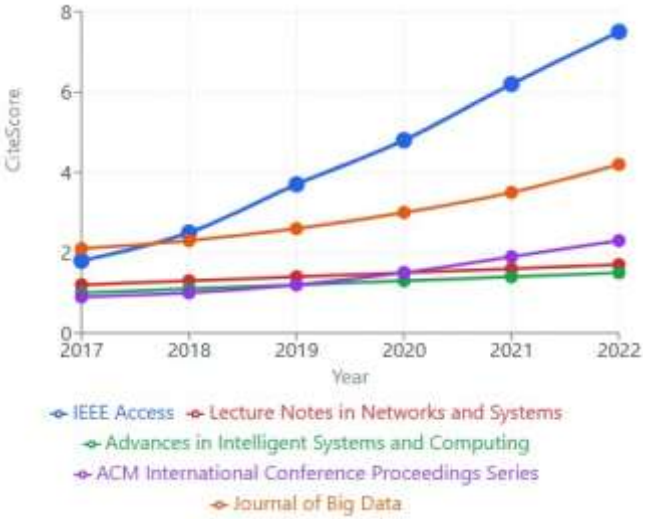


Fig. 6. Journals Cite score

B. Content Analysis

The following questions pertain specifically to the second stage, which is systematic review:

RQ4: How many credit card fraud detection articles utilize machine learning, deep learning, or both?

In Table 3, the analysis of credit card fraud detection reveals a substantial reliance on machine learning (ML) and deep learning (DL) techniques. Fig. 7 highlights the widespread use of these approaches, with 41 articles applying ML algorithms, 15 utilizing DL methods, and 19 integrating both. This distribution highlights the increasing adoption of advanced computational techniques in this domain, leveraging the respective strengths of each method. ML's adaptability and effectiveness in handling structured data are likely contributed to its broad application, while DL's ability to capture complex patterns within large, unstructured datasets is gaining prominence in addressing the evolving sophistication of fraud attempts.

Table 3: distribution of learning across papers

Learning	Paper	Paper Count
Machine Learning	[1], [2], [21]–[76]	41
Both	[4], [6], [21], [24], [27], [29], [31], [35], [37], [38], [40]–[42], [45], [50], [60], [63], [74], [77]–[90]	19
Deep Learning	[21]–[24], [27], [29], [35], [37], [38], [40]–[42], [45], [56], [60], [63], [74], [88]–[89]	15

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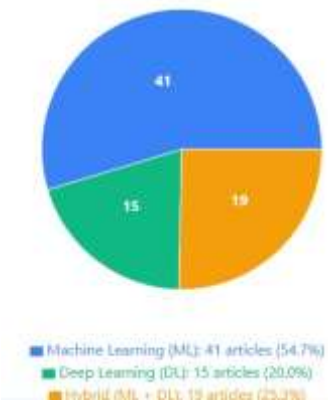


Fig. 7. Number of Articles Utilizing ML, DL, or Both

RQ5: What fraud detection algorithms are commonly used in this area?

The choice of specific algorithms reflects the diverse nature of fraud detection problems. Table 4 shows that the Random Forest is the most common (35 articles), followed by Logistic Regression (25) and SVM (24), often used for imbalanced datasets. Decision Trees (18) and KNN (14) are valued for interpretability. Advanced models like CNNs (13) and LSTMs capture complex patterns and temporal dependencies. Ensemble methods such as XGBoost (11) and AdaBoost (7), and ANNs (9) further enhance accuracy. Table IV summarizes the frequency of these models, highlighting the need for scalable, adaptive, and accurate fraud detection approaches.

Table 4: Distribution of algorithms across papers

Algorithm	Paper	Paper Count
Random Forest	[1]–[2], [21], [24]–[25], [27], [29], [32], [36]–[39], [41], [45]–[47], [49]–[51], [53], [55]–[56], [59], [63]–[64], [66]–[68], [70]–[71], [74]–[75], [89], [91]	35
Logistic Regression	[1], [21], [24]–[25], [29], [34], [36], [39], [41], [44], [46], [53], [55]–[56], [58]–[59], [65], [68]–[69], [71], [74], [76], [89], [91]	25
SVM	[1], [21]–[22], [29], [39], [40], [42], [47], [50], [52], [54], [55]–[56], [58], [60], [62], [65], [66], [69], [71]–[72], [74], [76], [89]	24
Decision Tree	[1], [21], [24], [27], [29]–[30], [34], [38], [39], [41], [53], [66], [71]–[72], [74], [76], [91]	18
KNN	[21], [29], [39], [42], [44], [47], [53], [55], [58], [65], [72], [76], [89], [91]	14
CNN	[31], [41]–[42], [44], [50], [63], [81], [82]–[84], [88], [89]	13
XGBoost	[2], [24]–[25], [30], [34], [46], [53], [55], [59], [70], [74]	11
ANN	[22]–[23], [30], [34], [44], [56], [64], [69]	9
LSTM	[6], [24], [35], [37], [40], [60], [63], [80], [88]	9
AdaBoost	[1]–[2], [29], [37], [49], [68], [71]	7

RQ6: How many studies discuss supervised, unsupervised, or semi-supervised techniques?

In Table 5, the analysis indicates a strong emphasis on supervised learning in credit card fraud detection research, with 58 articles employing this approach. A minority of studies (13) integrate both supervised and unsupervised techniques, while only 4 papers focus solely on unsupervised learning. Fig. 8 suggests a preference for supervised methods, likely due to the availability of labeled datasets, which facilitates model training and evaluation. The limited exploration of unsupervised approaches presents opportunities for novel research directions. Additionally, the inclusion of a single digital twin-related article underscores the current limited adoption of this emerging methodology.

Table 5: learning type distribution across papers

Learning Type	Paper	Paper count
Supervised	[1], [2], [4], [21]–[25], [27], [30]–[34], [36], [37], [39]–[44], [47], [48], [49], [53], [54], [55]–[56], [58]–[59], [62], [63]–[65], [66]–[72], [73]–[76], [79]–[84], [85], [86], [90], [91]	58
Both	[6], [29], [35], [38], [45]–[46], [50]–[52], [60], [74], [78], [89]	13
Unsupervised	[26], [28], [61], [77]	4
Digital Twin	[62]	1

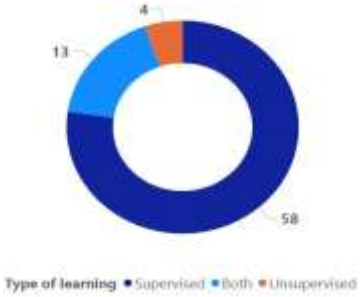


Fig. 8. Type of learning per paper

RQ8: How frequently are class imbalance issues addressed?

According to the analysis in Table 6, Random Under sampling is the most frequently used technique for addressing class imbalance, appearing in 11 papers. SMOTE follows closely, being utilized in 9 papers, highlighting its widespread adoption in balancing datasets. ADASYN is applied in 4 papers, while Feedback Mechanism is mentioned in 2 papers. Advanced approaches like Generative Adversarial Networks (GANs) and Incremental Learning are each referenced in 2 papers, demonstrating their growing significance in handling class imbalance. Hybrid methods, combining oversampling and under sampling strategies, appear in 1 paper, reflecting interest in integrated techniques. Additionally, emerging methods like Density-Oriented Fuzzy C-Means (DOFCM) and Value Clustering for Categorical Attributes (VCCA) are each mentioned in 1 paper, showcasing innovative solutions in the field.

Table 6: Resampling techniques and their paper distribution

Resampling Technique	Papers	Paper count
Random Under sampling	[6], [21], [34], [39], [42], [44]–[45], [47], [60], [75], [82]	11
SMOTE	[1], [34], [40], [56], [63], [68], [71], [73], [91]	9
ADASYN	[6], [29], [53], [82]	4
Random Oversampling	[34], [82], [85]	3
SMOTE-ENN	[37], [86]	2
Incremental learning	[42], [79]	2

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Adaboost	[43], [49]	2
Cluster-Based Under sampling	[44], [67]	2
Feedback mechanism	[45], [51]	2
BSMOTE	[40]	1
GAN	[40], [62]	2
MFGAN	[40]	1
SBS	[44]	1
Tomek links algorithm	[45]	1
GA in Feature aggregation	[21]	1
Oversampling followed by oversampling	[82]	1
Oversampling followed by under sampling	[82]	1
Under sampling followed by oversampling	[82]	1
Nearmiss	[76]	1
Minimize using class weights	[90]	1
Layer-Wise Relevance Propagation (LRP)	[84]	1
Local Interpretable Model-agnostic Explanations (LIME)	[84]	1
Shapley Additive Explanations (SHAP)	[84]	1
Density Oriented Fuzzy C Means	[73]	1
VCCA	[85]	1
KNN under sampling	[70]	1

RQ9: How frequently is the missing data problem addressed?

According to Table 7, only 3 out of 77 papers (approximately 4%) explicitly addressed the challenge of missing data, as seen in references [29], [2], and [35]. Conversely, a significant majority of 74 papers (about 96%) did not tackle this problem. This low frequency suggests that the missing data may be an underappreciated aspect within the field. Missing data can substantially affect the accuracy and reliability of machine learning models, as incomplete datasets can introduce bias and lead to unreliable results. The reasons for this oversight could include assumptions of complete data by researchers, the complexity of handling missing data, or a lack of awareness regarding its potential impact on model performance. Ignoring missing data can result in suboptimal model performance and potentially flawed conclusions. Therefore, future research should place greater emphasis on addressing missing data appropriately. Techniques such as mean imputation, regression imputation, or advanced methods like multiple imputation could enhance the robustness of AI models used in credit card fraud detection.

TABLE II. ADDRESSED MISSING DATA PROBLEM

Missing Data Problem	Paper	Paper count
Yes	[2], [29], [35]	3
No	[1, 2, 4, 6, 21-93] exclude: [29], [2], [35]	74

RQ10: How many articles incorporate optimization techniques?

Table 7 presents an overview of the 77 reviewed papers, revealing that 16 employed optimization algorithms, while 58 relied on standard or simpler methods. This distribution suggests that most studies did not require advanced optimization techniques, likely due to their relatively complete datasets with minimal missing values. The 16 papers that incorporated optimization algorithms may have faced more complex challenges, such as high-dimensional data or intricate feature interactions, necessitating more sophisticated approaches. In contrast, the remaining 58 papers may have opted for simpler methodologies due to the straightforward nature of their datasets or the adequacy of basic machine learning models for their specific use cases. As shown in Fig. 9, this variation in methodological choices highlights the diverse range of problems and data characteristics encountered in credit card fraud detection research.

Table 7: Optimization technique usage in papers

Optimization Usage	Paper	Paper count
Yes	[1], [4], [6], [33], [35], [39]–[41], [48]–[50], [72], [78], [79], [82], [88]	16
No	[1], [2], [21]–[30], [31]–[32], [34], [36]–[38], [42]–[47], [51]–[56], [58]–[62], [63]–[71], [73], [75]–[77], [80]–[81], [83]–[87], [90]–[91]	58

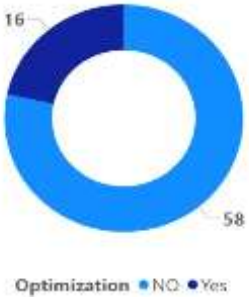


Fig. 9. Paper used optimization algorithms

RQ11: What are the top 10 AI-based optimization techniques used in credit card fraud detection, and how do they compare in terms of implementation frequency?

Table 8 outlines several AI-based optimization techniques used in credit card fraud detection, with the Genetic Algorithm being the most frequently used technique, appearing in four papers ([1], [48], [79], [49]). This method mimics natural selection and evolution to optimize complex problems, making it suitable for feature selection and parameter tuning. Particle Swarm Optimization (PSO) follows closely, applied in three papers ([35], [79], [72]), inspired by the social behavior of bird flocking or fish schooling and effective for optimizing multi-dimensional search spaces. The Adam optimizer, an adaptive learning rate optimization algorithm that combines the advantages of AdaGrad and RMSProp, has been utilized in two papers

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([82], [6]) for efficient and robust performance in training machine learning models. The distribution of these top three optimization algorithms is visually represented in Fig 10**Error! Reference source not found.** Other less frequently used techniques include AGTO, Ant Colony Optimization, Bat Optimization Algorithm, COA, CoatiOA, Competitive Swarm Optimization (CSO), Differential Evolution Feature Selection (DEFS), Genetic Algorithm Feature Selection (GAFS), Grid Search, Modular Neural Network, MSGD, Neural Aggregate Generator (NAG), Polynomial Self-Learning (PSLPSO), Random Forest hyperparameter tuning, Rock Hyrax Swarm Optimization (RHSO), and SGD. These methods address challenges such as high-dimensional data handling, model accuracy improvement, and computational efficiency enhancement in credit card fraud detection.

Table 8: optimization algorithms

Optimization algorithm	Paper	Paper count
Genetic Algorithm	[1], [48], [79], [49]	4
Particle Swarm	[35], [79], [72]	3
Adam	[82], [6]	**2**
AGTO	[33]	1
Ant colony optimization	[49]	1
Bat Optimization Algorithm	[41]	1
COA	[33]	1
CoatiOA	[33]	1
Competitive Swarm Optimization	[4]	1
Differential Evolution Feature Selection (DEFS)	[72]	1
Genetic Algorithm Feature Selection	[72]	1
Grid Search	[40]	1
Modular Neural Network	[78]	1
MSGD	[82]	1
Neural Aggregate Generator (NAG)	[88]	1
Polynomial Self-Learning (PSLPSO)	[39]	1
Random forest hyperparameter tuning	[49]	1
Rock Hyrax Swarm Optimization (RHSO)	[72]	1
SGD	[82]	1

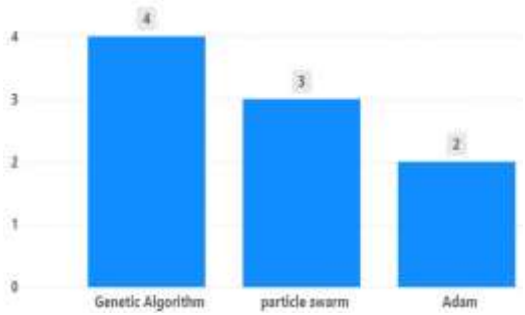


Fig. 10. Top 3 used optimization algorithms

RQ12: What gaps remain in this area?

The primary gaps in credit card fraud detection research include the limited integration of large language models (LLMs), despite their growing application in other fields. Among the reviewed papers, none utilized LLMs, highlighting a missed opportunity to leverage their potential for improving fraud detection through enhanced feature extraction, context understanding, and anomaly detection. Additionally, while missing data is a common challenge in financial datasets, only 3 out of 77 papers addressed this issue, indicating the need for more robust handling techniques. The underutilization of LLMs and the inadequate attention to missing data suggest that researchers have yet to fully leverage these tools to enhance model performance and practical applicability. Future studies should focus on integrating LLMs into fraud detection frameworks and developing advanced methods for managing incomplete or noisy data. Addressing these gaps could lead to more accurate and reliable AI-driven solutions, better equipped to handle the complexities of realworld financial transactions.

VIII. Conclusion

This systematic review has moved beyond a mere synthesis of current practices, offering a critical AI-centric analysis of machine learning and deep learning approaches in credit card fraud detection. We underscore that while traditional supervised learning and established algorithms remain prevalent, the field urgently requires a strategic shift towards addressing fundamental AI challenges posed by the dynamic and adversarial nature of financial fraud. Our comprehensive roadmap highlights critical future research avenues: the integration of sophisticated AI optimization strategies, the development of robust AI techniques for missing data and concept drift, and the exploration of novel AI paradigms such as Large Language Models (LLMs) and Graph Neural Networks (GNNs). Crucially, the future of AI in fraud detection hinges advancements in AI and building inherently robust models against adversarial attacks. By committing to this research agenda, the AI community can collaboratively develop the next generation of intelligent, adaptive, and trustworthy fraud detection systems, thereby fortifying global financial ecosystems against increasingly sophisticated threats.

References

[1] I. D. Mienye and Y. Sun, "A machine learning method with hybrid feature selection for improved credit card fraud detection," Applied Sciences, vol. 13, no. 12, 2023.

[2] D. P. Prabha and C. V. Priscilla, "Probabilistic XGBoost threshold classification with autoencoder for credit card fraud detection," International Journal on Recent and Innovation Trends in Computing and Communication, vol. 11, no. 8s, pp. 528-537, 2023.

[3] O. A. Bello, A. F. Ogundipe, O. K. Folake, Z. Budale, and O. E. Ejiofor, "Enhancing cyber financial fraud detection using deep learning techniques: A study on neural networks and anomaly detection," 2022.

[4] E. Rawashdeh et al., "Efficient credit card fraud detection using evolutionary hybrid feature selection and random weight networks," International Journal of Data and Network Science, vol. 8, no. 1, pp. 463472, 2024.

[5] S. N. Kalid et al., "Detecting frauds and payment defaults on credit card data inherited with imbalanced class distribution and overlapping class problems: A systematic review," IEEE Access, vol. 12, pp. 23636-23652, 2024.

[6] S. S. Sulaiman, I. Nadher, and S. M. Hameed, "Credit card fraud detection using improved deep learning models," Computers, Materials & Continua, vol. 78, no. 1, pp. 1049-1069, 2024.

[7] R. A. Derrig, "Insurance fraud," Journal of Risk and Insurance, vol. 69, no. 3, pp. 271-287, 2002.

[8] M. L. Bhasin, "The role of technology in combatting bank frauds: Perspectives and prospects," ECOFORUM, 2016.

[9] W. Hilal, S. A. Gadsden, and J. Yawney, "Financial fraud: A review of anomaly detection techniques and recent advances," Expert Systems with Applications, vol. 193, 2022.

[10] L. Popescu, "Addressing tax fraud in the tax literature," Scientific Bulletin – Economic Sciences, 2021.

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[11] A. T. Carswell and D. C. Bachtel, "Mortgage fraud: A risk factor analysis of affected communities," Crime, Law and Social Change, vol. 52, no. 4, pp. 347-364, 2009.
[12] C. Janiesch, P. Zschech, and K. Heinrich, "Machine learning and deep learning," Electronic Markets, vol. 31, no. 3, pp. 685-695, 2021.
[13] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," Nature, vol. 521, no. 7553, pp. 436-444, 2015.
[14] B. Reddy and R. Fields, "From past to present – A comprehensive technical review of rule-based expert systems from 1980 - 2021," in Proc. ACM Southeast Conference, 2022, pp. 167-172.
[15] M.-H. Guo et al., "Attention mechanisms in computer vision: A survey," Computational Visual Media, vol. 8, no. 3, pp. 331-368, 2022.
[16] H. B. Essel et al., "The impact of a virtual teaching assistant (chatbot) on students' learning in Ghanaian higher education," International Journal of Educational Technology in Higher Education, vol. 19, no. 1, 2022.
[17] M. N. Ashtiani and B. Raahemi, "Intelligent fraud detection in financial statements using machine learning and data mining: A systematic literature review," IEEE Access, vol. 10, pp. 72504-72525, 2022.
[18] M. Sánchez-Aguayo, L. Urquiza-Aguiar, and J. Estrada-Jiménez, "Fraud detection using the fraud triangle theory and data mining techniques: A literature review," Computers, vol. 10, no. 10, 2021.
[19] E. A. L. Marazqah Btoush et al., "A systematic review of literature on credit card cyber fraud detection using machine and deep learning," PeerJ Computer Science, vol. 9, p. e1278, 2023.
[20] R. M. Dantas et al., "Systemic acquired critique of credit card deception exposure through machine learning," Journal of Open Innovation: Technology, Market, and Complexity, vol. 8, no. 4, 2022.
[21] M. Seera et al., "An intelligent payment card fraud detection system," Annals of Operations Research, pp. 1-23, 2021.
[22] O. Ndama, I. Bensassi, and E. M. En-naimi, "Innovative credit card fraud detection: A hybrid model combining artificial neural networks and support vector machines," IAES International Journal of Artificial Intelligence (IJ-AI), vol. 13, no. 3, 2024.
[23] K. Sreekanth et al., "Prevention of credit card fraud transaction using GA feature selection for web-based application," Indonesian Journal of Electrical Engineering and Computer Science, vol. 35, no. 3, 2024.
[24] I. D. Mienye and T. G. Swart, "A hybrid deep learning approach with generative adversarial network for credit card fraud detection," Technologies, vol. 12, no. 10, 2024.
[25] C. Yan et al., "Enhancing credit card fraud detection through adaptive model optimization," in Proc. IEEE 7th International Conference on Big Data and Artificial Intelligence (BDAI), 2024, pp. 49-54.
[26] M. Katsouda et al., "Mining association of outliers in time series," in Recent Advancements in Tourism Business, Technology and Social Sciences, 2024, pp. 433-444.
[27] R. K. L. Kennedy et al., "Synthesizing class labels for highly imbalanced credit card fraud detection data," Journal of Big Data, vol. 11, no. 1, 2024.
[28] H. Oh and D. K. Park, "Quantum support vector data description for anomaly detection," Machine Learning: Science and Technology, vol. 5, no. 3, 2024.
[29] R. Ku et al., "A machine learning based approach for the fraud detection in imbalanced credit card transaction dataset," International Journal of Electronics and Communication Engineering, vol. 11, no. 8, pp. 244-259, 2024.
[30] B. S. Bala, P. P. Yadav, and M. R. Reddy, "An intelligent approach to detect and predict online fraud transaction using XGBoost algorithm," Indonesian Journal of Electrical Engineering and Computer Science, vol. 35, no. 3, 2024.
[31] S. Al Balawi and N. Aljohani, "Credit-card fraud detection system using neural networks," The International Arab Journal of Information Technology, vol. 20, no. 2, 2023.
[32] K. K. Renganathan et al., "Credit card fraud detection with advanced graph based machine learning techniques," Indonesian Journal of Electrical Engineering and Computer Science, vol. 35, no. 3, 2024.
[33] M. A. Salam, D. L. E.-B., K. M. Fouad, and M. S. E. Elsayed, "Enhancing fraud detection in credit card transactions using optimized federated learning model," International Journal of Advanced Computer Science and Applications, 2024.
[34] P. Gupta, A. V., M. R. Khan, R. Ahmed, M. Shuaib, and S. Alam, "Unbalanced credit card fraud detection data: A machine learningoriented comparative study of balancing techniques," Procedia Computer Science, 2023.
[35] P. K. Sree, V. Kakulapati, and B. Indira, "A novel method of fraud detection of credit cards by fuzzy, LSTM, and PSO optimization," International Journal on Recent and Innovation Trends in Computing and Communication, vol. 11, no. 7, pp. 229-234, 2023.
[36] M. Valavan and S. Rita, "Predictive-analysis-based machine learning model for fraud detection with boosting classifiers," Computer Systems Science and Engineering, vol. 45, no. 1, pp. 231-245, 2023.
[37] I. D. Mienye and Y. Sun, "A deep learning ensemble with data resampling for credit card fraud detection," IEEE Access, vol. 11, pp. 30628-30638, 2023.
[38] R. Gupta et al., "AI-enhanced data mining for fraud detection in financial transactions," in Proc. 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL), 2024, pp. 244-249.
[39] A. Mniai, M. Tarik, and K. Jebari, "A novel framework for credit card fraud detection," IEEE Access, vol. 11, pp. 112776-112786, 2023.
[40] Y. Xie et al., "A credit card fraud detection model based on multi-feature fusion and generative adversarial network," Computers, Materials & Continua, vol. 76, no. 3, pp. 2707-2726, 2023.
[41] H. Sikkandar et al., "An intelligent approach for anomaly detection in credit card data using bat optimization algorithm," Inteligencia Artificial, vol. 26, no. 72, pp. 202-222, 2023.
[42] X. Feng and S.-K. Kim, "Novel machine learning based credit card fraud detection systems," Mathematics, vol. 12, no. 12, 2024.
[43] V. H. Khang, C. T. A., and N. D. Thuan, "Detecting fraud transaction using Ripper algorithm combines with ensemble learning model," International Journal of Advanced Computer Science and Applications, 2023.
[44] H. Ahmad et al., "Class balancing framework for credit card fraud detection based on clustering and similarity-based selection (SBS)," International Journal of Information Technology, vol. 15, no. 1, pp. 325333, 2023.
[45] A. Ruchay et al., "The imbalanced classification of fraudulent bank transactions using machine learning," Mathematics, vol. 11, no. 13, 2023.
[46] J. L. Leevy et al., "Investigating the effectiveness of one-class and binary classification for fraud detection," Journal of Big Data, vol. 10, no. 1, 2023.
[47] A. R. Khalid et al., "Enhancing credit card fraud detection: An ensemble machine learning approach," Big Data and Cognitive Computing, vol. 8, no. 1, 2024.
[48] A.-W. K. Al-Faqeh et al., "Credit card fraud detection via integrated account and transaction submodules," Arabian Journal for Science and Engineering, vol. 46, no. 10, pp. 10023-10031, 2021.
[49] V. S. S. Karthik, A. Mishra, and U. S. Reddy, "Credit card fraud detection by modelling behaviour pattern using hybrid ensemble model," Arabian Journal for Science and Engineering, vol. 47, no. 2, pp. 1987-1997, 2021.
[50] S. Ingole et al., "Service-based credit card fraud detection using Oracle SOA Suite," SN Computer Science, vol. 2, no. 3, 2021.
[51] U. Kirithika, S. K. S. Raja, C. J. Raman, and V. Balaji, "A novel fraud detection scheme for credit card usage employing random forest algorithm combined with feedback mechanism," Journal of Information Technology Management, 2021.
[52] I. Sadgali, N. Sael, and F. Benabbou, "Human behavior scoring in credit card fraud detection," IAES International Journal of Artificial Intelligence, vol. 10, no. 3, 2021.
[53] A. Priyadarshini et al., "Fraudulent credit card transaction detection using soft computing techniques," Indonesian Journal of Electrical Engineering and Computer Science, vol. 23, no. 3, 2021.
[54] K. Poongodi and D. Kumar, "Support vector machine with information gain based classification for credit card fraud detection system," The International Arab Journal of Information Technology, vol. 18, no. 2, 2021.
[55] N. R. Dzakiyullah, A. Pramuntadi, and A. K. Fauziyyah, "Semisupervised classification on credit card fraud detection using autoencoders," Journal of Applied Data Sciences, 2021.
[56] P. Sharma et al., "Machine learning model for credit card fraud detection - A comparative analysis," The International Arab Journal of Information Technology, 2021.
[57] P. Chatterjee, D. Das, and D. B. Rawat, "Digital twin for credit card fraud detection: Opportunities, challenges, and fraud detection advancements," Future Generation Computer Systems, vol. 158, pp. 410-426, 2024.
[58] S. G. Deshpande, G. P. Gawali, R. B. Ghayalkar, S. M. Chavan, J. H. Meshram, and D. R. Tripathi, "Machine learning for fraud detection: Comparative analysis of algorithms and performance evaluation," Nanotechnology Perceptions, 2024.
[59] O. Ogundile et al., "Credit card fraud: Analysis of feature extraction techniques for ensemble hidden Markov model prediction approach," Applied Sciences, vol. 14, no. 16, 2024.
[60] S. Saad, I. Nadher, and S. M. Hameed, "Credit card fraud detection challenges and solutions: A review," Iraqi Journal of Science, pp. 22872303, 2024.
[61] Y. Huang et al., "Interpretable single-dimension outlier detection (ISOD): An unsupervised outlier detection method based on quantiles and skewness coefficients," Applied Sciences, vol. 14, no. 1, 2023.
[62] B. Alshawi, "Comparison of SVM kernels in credit card fraud detection using GANs," International Journal of Advanced Computer Science and Applications, 2024.
[63] T. Baabdullah et al., "Efficiency of federated learning and blockchain in preserving privacy and enhancing the performance of credit card fraud detection (CCFD) systems," Future Internet, vol. 16, no. 6, 2024.
[64] A. Muaz, M. Jayabalan, and V. Thiruchelvam, "A comparison of data sampling techniques for credit card fraud detection," International Journal of Advanced Computer Science and Applications, 2020.

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[65] N. Rosley, G. K. Tong, K. H. Ng, S. N. Kalid, and K. C. Khor, "Autoencoders with reconstruction error and dimensionality reduction for credit card fraud detection," Journal of System and Management Sciences, 2022.
[66] V. Plakandaras et al., "Credit card fraud detection with automated machine learning systems," Applied Artificial Intelligence, vol. 36, no. 1, 2022.
[67] W. Li, C.-S. Wu, and S.-M. Ruan, "CUS-RF-based credit card fraud detection with imbalanced data," Journal of Risk Analysis and Crisis Response, vol. 12, no. 3, 2022.
[68] S. Saraf and A. Phakatkar, "Credit card fraud detection using hybrid ensemble model: Bagging and boosting approaches for data imbalance," International Journal of Advanced Computer Science and Applications, 2022.
[69] S. Khan, A. Alourani, B. Mishra, A. Ali, and M. Kamal, "Developing a credit card fraud detection model using machine learning approaches," International Journal of Advanced Computer Science and Applications, 2022.
[70] N. S. Alfaiz and S. M. Fati, "Enhanced credit card fraud detection model using machine learning," Electronics, vol. 11, no. 4, 2022.
[71] M. R. Baker, Z. N. Mahmood, and E. H. Shaker, "Ensemble learning with supervised machine learning models to predict credit card fraud transactions," Revue d'Intelligence Artificielle, vol. 36, no. 4, pp. 509518, 2022.
[72] B. K. Padhi et al., "RHSOFS: Feature selection using the rock hyrax swarm optimization algorithm for credit card fraud detection system," Sensors, vol. 22, no. 23, 2022.
[73] K. Upadhyay, P. Kaur, and R. Sachdeva, "Fast and robust fuzzy-based hybrid data-level method to handle class imbalance," International Journal of Advanced Computer Science and Applications, 2022.
[74] G. Schrijver, D. K. Sarmah, and M. El-hajj, "Automobile insurance fraud detection using data mining: A systematic literature review," Intelligent Systems with Applications, vol. 21, 2024.
[75] G. M. Paldino et al., "The role of diversity and ensemble learning in credit card fraud detection," Advances in Data Analysis and Classification, pp. 1-25, 2022.
[76] F. Jeribi, "A comprehensive machine learning framework for anomaly detection in credit card transactions," International Journal of Advanced Computer Science & Applications, 2024.
[77] F. Adesola et al., "Design and implementation of a web-based credit card fraud detection system," in Proc. International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG), 2024, pp. 1-10.
[78] F. O. Aghware, R. E. Y., P. O. Ejeh, C. C. Odiakaose, F. U. Emordi, and A. A. Ojugo, "DeLClustE: Protecting users from credit card fraud transaction via the deep learning cluster ensemble," International Journal of Advanced Computer Science and Applications, vol. 14, no. 6, 2023.
[79] B. Lebichot et al., "Incremental learning strategies for credit cards fraud detection," International Journal of Data Science and Analytics, vol. 12, no. 2, pp. 165-174, 2021.
[80] I. Sadgali, N. Sael, and F. Benabbou, "Bidirectional gated recurrent unit for improving classification in credit card fraud detection," Indonesian Journal of Electrical Engineering and Computer Science, vol. 21, no. 3, 2021.
[81] N. Mqadi, N. Naicker, and T. Adeliyi, "A SMOTE based oversampling data-point approach to solving the credit card data imbalance problem in financial fraud detection," International Journal of Computing and Digital Systems, vol. 10, no. 1, pp. 277-286, 2021.
[82] M. Abdul Salam et al., "Federated learning model for credit card fraud detection with data balancing techniques," Neural Computing and Applications, vol. 36, no. 11, pp. 6231-6256, 2024.
[83] V. G. Krishnan et al., "Development of deep learning based intelligent approach for credit card fraud detection," International Journal on Recent and Innovation Trends in Computing and Communication, vol. 10, no. 12, pp. 133-139, 2022.
[84] I. Ullah et al., "Explaining deep learning models for tabular data using layer-wise relevance propagation," Applied Sciences, vol. 12, no. 1, 2021.
[85] E. M. Carneiro et al., "High-cardinality categorical attributes and credit card fraud detection," Mathematics, vol. 10, no. 20, 2022.
[86] V. Narayan and S. Ganapathisamy, "Hybrid sampling and similarity attention layer in bidirectional long short term memory in credit card fraud detection," International Journal of Intelligent Engineering and Systems, vol. 15, no. 6, pp. 35-44, 2022.
[87] B. Kasasbeh, B. Aldabaybah, and H. Ahmad, "Multilayer perceptron artificial neural networks-based model for credit card fraud detection," Indonesian Journal of Electrical Engineering and Computer Science, vol. 26, no. 1, 2022.
[88] K. Ghosh Dastidar et al., "NAG: Neural feature aggregation framework for credit card fraud detection," Knowledge and Information Systems, vol. 64, no. 3, pp. 831-858, 2022.
[89] D. Hove, O. Olugbara, and A. Singh, "Bibliometric analysis of recent trends in machine learning for online credit card fraud detection," Journal of Scientometric Research, vol. 13, no. 1, pp. 43-57, 2024.
[90] O. Ndama, I. Bensassi, and E. M. En-Naimi, "Optimizing credit card fraud detection: A deep learning approach to imbalanced datasets," International Journal of Electrical and Computer Engineering (IJECE), vol. 14, no. 4, 2024.
[91] B. Vihurskyi, "Credit card fraud detection with XAI: Improving interpretability and trust," in Proc. Third International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE), 2024, pp. 1-6.
[92] R. Alothman, H. A. Talib, and M. S. Mohammed, "Fraud detection under the unbalanced class based on gradient boosting," Eastern-European Journal of Enterprise Technologies, vol. 2, no. 2(116), pp. 6-12, 2022.
[93] I. D. Mienye and N. Jere, "Deep learning for credit card fraud detection: A review of algorithms, challenges, and solutions," IEEE Access, vol. 12, pp. 96893-96910, 2024.

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